



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

ECHO PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on routing, middleware, binding, JWT auth, and testing

TOTAL: 10 QUESTIONS

Q1 What is Echo and how does it compare to Gin?

Threat Model

Q6 What does echo.Context provide for reading request data?

Observability

Q2 How do you apply middleware globally vs to a route group in Echo?

Architecture

Q7 How do you implement JWT authentication in Echo?

Lifecycle

Q3 How does request binding and validation work in Echo?

Configuration

Q8 What are Echo's response helpers and what do they return?

Compliance

Q4 How do Echo Groups help organize routes?

Hardening

Q9 How do you handle WebSocket connections in Echo?

Testing

Q5 How does Echo's custom error handler work?

SSO / IdP

Q10 How do you test Echo handlers without starting an HTTP server?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Both use radix-tree routing. Echo provides

Mitigation

Both use radix-tree routing. Echo provides built-in middleware (Logger, Recover, CORS, JWT), automatic TLS via Let's Encrypt, and HTTP/2 support out of the box. Its echo.Context API is arguably more ergonomic with typed header/cookie helpers.

A6 c.Param('id') reads path params,

Observability

c.Param('id') reads path params, c.QueryParam('q') reads query strings, c.FormValue('field') reads form fields, c.Request().Header.Get('X-API-Key') reads headers, and c.Get/Set passes values through middleware to handlers via the request context.

A2 e.Use(middleware.Logger()) applies globally to all

Primitives

e.Use(middleware.Logger()) applies globally to all routes. `g := e.Group('/admin', AdminMiddleware)` applies to the group only. Individual routes: `e.GET('/profile', handler, AuthMiddleware)`. Middleware functions have signature `func(next echo.HandlerFunc) echo.HandlerFunc`.

A7 Register

Automation

Register `echojwt.WithConfig(echojwt.Config{SigningKey: []byte(secret)})` as middleware on a Group. It reads the Bearer token, validates the signature, and populates the `jwt.Token` in the context. Retrieve claims with `c.Get('user').(*jwt.Token).Claims`.

A3 c.Bind(&req) reads JSON body, form values,

Hardening

`c.Bind(&req)` reads JSON body, form values, or query params into a struct based on tags. Echo does not validate by default. Register a validator (`e.Validator = &MyValidator{}`) that implements `Validate(i interface{}) error` and call `c.Validate(&req)` explicitly.

A8 c.JSON(200, obj) marshals obj and sets

Governance

`c.JSON(200, obj)` marshals `obj` and sets `Content-Type` `application/json`. `c.XML`, `c.String`, `c.HTML`, `c.File`, `c.Stream` cover other content types. `c.NoContent(204)` sends an empty response. `c.Redirect(301, url)` sets `Location` header and status.

A4 g := e.Group('/api/v1') prefixes all routes

Gotchas

`g := e.Group('/api/v1')` prefixes all routes in the group. Apply shared middleware once: `g.Use(JWTMiddleware)`. Create sub-groups: `admin := g.Group('/admin', AdminOnly)`. This mirrors `Express.Router` and `NestJS Module` organization patterns.

A9 Use github.com/gorilla/websocket inside an

Assessment

Use `github.com/gorilla/websocket` inside an Echo handler. Register a standard `echo.HandlerFunc`, upgrade the connection with `upgrader.Upgrade(c.Response(), c.Request(), nil)`, then loop on `conn.ReadMessage()` and `conn.WriteMessage()` for full-duplex communication.

A5 Set e.HTTPErrorHandler = func(err error, c

Federation

Set `e.HTTPErrorHandler = func(err error, c echo.Context)`. Check if `err` is `*echo.HTTPError` for status/message; otherwise default to 500. This centralizes error formatting, allowing you to return consistent JSON error shapes across the entire application.

A10 req := httptest.NewRequest(http.MethodGet,

Optimization

`req := httptest.NewRequest(http.MethodGet, '/users/1', nil)`; `rec := httptest.NewRecorder()`; `c := e.NewContext(req, rec)`; `handler(c)`. Access `rec.Code` and `rec.Body.String()` for assertions. No listening port is required.